

INTRODUCTION

Financial crises have been a recurring challenge throughout history, profoundly shaping economies, societies, and political landscapes. From the Great Depression in 1929 to the Global Financial Crisis of 2008, these events have consistently revealed weak spots in financial systems. They also remind us of the importance of having tools in place to predict and manage such disruptions. In today's hyper-connected world, an economic shock in one region can quickly spread across the globe, making it essential to adopt proactive strategies to recognize and address potential financial crises before they escalate.

Globalization has brought the world's economies closer together, creating both opportunities and risks. By removing trade barriers, advancing technology, and enabling open capital flows, we've built a tightly interconnected system where the fortunes of one nation often depend on others. While this integration has fueled innovation and growth, it has also left economies more exposed to instability. The Asian Financial Crisis of 1997 showed how financial troubles can cross borders, while the 2008 crisis revealed how vulnerabilities in the U.S. economy could trigger global repercussions. These events underline the need for effective systems to identify and tackle emerging threats.

To spot early warning signs of financial instability, we focus on monitoring a range of critical indicators. These metrics give us insights into the health of financial systems and broader economic trends:

1. **Gold Prices:** Gold is seen as a safe-haven asset during uncertain times. When investors lose confidence in traditional markets, they often turn to gold, driving its price up. A sharp rise in gold prices can serve as a red flag for declining confidence in the financial system.
2. **Bond Yields (U.S. and China):** Bond yields are a reflection of economic expectations, interest rate trends, and credit conditions. U.S. Treasury yields act as a global benchmark, while China's bond yields provide insight into the economic dynamics of one of the world's largest economies. Large movements in these yields can indicate stress in credit markets or global liquidity shifts, signaling potential economic challenges.
3. **Stock Market Indices (U.S., Germany, Japan, India):** Stock market indices offer real-time insights into investor sentiment and economic performance. By analyzing indices from leading economies, we can track global trends and their ripple effects. A steep drop in indices like Germany's DAX or Japan's Nikkei could signal broader economic issues that might eventually impact U.S. markets.
4. **U.S. Mortgage Rates:** The housing market has often played a central role in financial crises, as seen in 2008. Changes in mortgage rates affect consumer spending and household debt levels. Significant rate increases can put pressure on household budgets and dampen borrowing, which may hint at broader economic trouble.

5. **Dollar Index:** The U.S. dollar's strength is a key indicator of global trade, liquidity, and capital flows. A stronger dollar can strain emerging markets with dollar-denominated debt, while a weaker dollar might signal investor caution or waning confidence in the economy. These shifts can have widespread implications for both domestic and global markets.

Each of these indicators paints a part of the bigger picture:

- Rising gold prices and shifting bond yields reflect market stress and risk perception.
- Stock market indices highlight changes in economic confidence across key regions.
- Mortgage rates provide insight into the financial health of households.
- The dollar index reveals the U.S.'s position in global trade and financial dynamics.

By piecing these elements together, we're building a comprehensive framework to identify early signs of financial crises. This holistic approach considers both domestic and international factors, ensuring a balanced perspective that's crucial in today's interconnected financial landscape. With this framework, we aim to anticipate potential risks, safeguard our clients, and contribute to a stronger, more resilient financial system.

LITERATURE REVIEW

(Demyanyk & Hasan, 2009) The prediction of financial crises has been a critical area of research, integrating both econometric and machine learning techniques to improve forecasting accuracy. Existing literature highlights the multifaceted nature of financial crises, emphasizing the role of macroeconomic variables, banking sector vulnerabilities, and external shocks. Early studies, such as those by Demirguc-Kunt and Detragiache (1998), identify key predictors, including GDP growth, inflation, and real interest rates, that contribute to banking instability.

Operations research models have also played a significant role in crisis prediction. Since Altman's (1968) development of the Z-score model for bankruptcy prediction, research has expanded to include artificial intelligence-based approaches. Neural networks, support vector machines, and fuzzy clustering techniques have shown promise in identifying patterns leading to financial distress. Studies by Celik and Karatepe (2007) demonstrate that artificial neural networks can effectively forecast banking sector risks by analyzing non-performing loans, capital adequacy, and profitability metrics.

Moreover, early warning systems (EWS) are widely used by central banks to monitor financial stability. Empirical comparisons, such as those by Davis and Karim (2008), suggest that while logistic regression models perform well at a global level, country-specific models using signal extraction techniques yield better localized predictions. The integration of these models offers a

robust framework for anticipating financial crises, supporting policymakers in implementing timely interventions.

(El-Shagi et al., 2012) The literature on financial crisis prediction explores a range of methodologies, including early warning systems, econometric modeling, and machine learning techniques. The signals approach, pioneered by Kaminsky and Reinhart (1999), remains widely used, identifying economic indicators that signal potential crises, such as exchange rate fluctuations, declining reserves, and excessive credit growth. Other models, including binary-choice regression and Markov-switching techniques, provide alternative forecasting methods but often require restrictive assumptions regarding crisis probability and indicator relationships.

Recent research highlights the integration of artificial intelligence and big data analytics in crisis prediction. Neural networks, support vector machines, and deep learning algorithms have shown potential in capturing complex nonlinear relationships among economic variables. Studies by Celik and Karatepe (2007) demonstrate the effectiveness of machine learning techniques in predicting banking sector risks. Early warning systems developed by institutions like the International Monetary Fund (IMF) and central banks incorporate composite indicators to improve accuracy. These systems aggregate macroeconomic, financial, and external sector variables to provide a more comprehensive risk assessment. Empirical research suggests that while global models offer broad insights, country-specific models often yield more accurate crisis predictions.

The endstate of financial crisis prediction underscores the need for interdisciplinary approaches, combining traditional econometric models with machine learning advancements to enhance predictive accuracy and policy response.

(Aydin & Cavdar, 2015) The literature on financial crisis prediction integrates diverse methodologies, including early warning systems, econometric modeling, and artificial intelligence techniques. Traditional approaches, such as the signals model by Kaminsky and Reinhart (1999), identify macroeconomic indicators—exchange rate fluctuations, declining reserves, and excessive credit growth—as precursors to financial instability. Logit regression and Markov-switching models have been widely used but often suffer from rigid assumptions and limited adaptability.

Recent studies emphasize machine learning techniques, particularly Artificial Neural Networks (ANNs), as effective tools for financial crisis forecasting. Aydin and Cavdar (2015) applied ANN models to the Turkish economy and found that multilayer feedforward networks provide superior predictive accuracy over conventional statistical models. The ANN model's ability to learn complex, nonlinear relationships enables it to capture subtle financial distress signals missed by traditional methods.

Early warning systems developed by global financial institutions further enhance crisis prediction by aggregating macroeconomic, financial, and external sector indicators. While global models

offer broad insights, country-specific models improve predictive accuracy by incorporating localized risk factors. The integration of traditional econometric approaches with AI-driven models enhances the robustness of financial crisis prediction, providing valuable insights for policymakers and investors in mitigating systemic risks.

(Ricardo Laborda) The literature on financial crisis prediction explores various methodologies, including early warning systems, econometric models, and machine learning techniques. Traditional models, such as those developed by Kaminsky and Reinhart (1999), utilize macroeconomic indicators like exchange rate fluctuations, credit growth, and foreign reserves to anticipate crises. Econometric approaches, such as logit regression and Markov-switching models, have also been employed but often face limitations due to restrictive assumptions about crisis probability and indicator relationships.

Recent studies emphasize the role of volatility spillovers in predicting financial distress. Laborda and Olmo (2020) examine sectoral volatility connectedness and find that shocks from Banking & Insurance, Energy, and Technology significantly impact market-wide stability. Their research demonstrates that an increase in volatility transmission among economic sectors is a reliable indicator of impending financial crises. Utilizing network connectivity methods, they establish that heightened spillovers precede major crises, such as the 2008 financial meltdown and the COVID-19 market turmoil.

Furthermore, advancements in machine learning have improved predictive accuracy. Random forests and neural networks capture nonlinear relationships and detect early signals of financial instability. These AI-driven models complement traditional approaches, offering policymakers and investors robust frameworks for crisis mitigation. The integration of econometric and AI-based techniques continues to refine financial crisis prediction methodologies.

(Fernández-Arias et al., 2017) The literature on financial crisis prediction has evolved from traditional econometric models to more advanced machine learning techniques. Early warning systems, such as those developed by Kaminsky and Reinhart (1999), rely on macroeconomic indicators like exchange rate fluctuations, credit growth, and declining reserves to anticipate crises. Traditional models, including logit regression and Markov-switching approaches, have been widely used but often fail to capture complex, nonlinear relationships between financial variables.

Recent studies emphasize the role of artificial intelligence in crisis forecasting. Fernández-Arias et al. (2018) propose a hybrid multi-classification model that integrates extreme learning machines (ELM) with the Synthetic Minority Over-sampling Technique (SMOTE) to improve the prediction of financial distress in OECD banks. Their model successfully classifies banks into three categories based on financial health, demonstrating higher accuracy compared to traditional techniques.

Furthermore, network connectivity approaches highlight the importance of volatility spillovers in predicting systemic financial instability. These methods identify critical interdependencies between financial institutions, providing insights into how shocks propagate across sectors. The integration of AI-driven models with traditional econometric frameworks enhances financial crisis prediction, enabling policymakers and investors to implement proactive risk mitigation strategies.

(Bussière & Fratzscher, 2002) The literature on financial crisis prediction has evolved from traditional econometric models to advanced machine learning and network-based approaches. Early studies, such as Kaminsky and Reinhart (1999), used macroeconomic indicators—including exchange rate fluctuations, credit growth, and declining reserves—to signal impending crises. However, these models faced limitations due to their rigid assumptions and inability to capture nonlinear relationships.

Recent research has introduced more sophisticated early warning systems (EWS) to enhance prediction accuracy. Bussière and Fratzscher (2002) developed a multinomial logit EWS model that addresses the "post-crisis bias" by distinguishing between pre-crisis, crisis, and recovery periods. Their approach significantly improves forecasting by incorporating a broader range of economic and financial indicators.

Moreover, machine learning techniques, such as neural networks and random forests, have been increasingly applied to crisis prediction. These models excel in detecting complex patterns in financial data and can complement traditional econometric approaches. Additionally, financial network analysis, which examines sectoral volatility spillovers, has proven valuable in identifying systemic risks before they escalate.

By integrating traditional econometric models with AI-driven techniques, modern research continues to refine financial crisis prediction, offering policymakers and investors more robust tools for risk assessment and crisis prevention.

(Bluwstein et al., 2021) The literature on financial crisis prediction has evolved significantly, incorporating traditional econometric models and advanced machine learning techniques. Early studies, such as Kaminsky and Reinhart (1999), emphasized macroeconomic indicators like exchange rate fluctuations, credit growth, and interest rate differentials as key predictors. Logistic regression and Markov-switching models have been widely applied but often struggle with capturing complex, nonlinear relationships between financial variables.

Recent research highlights the effectiveness of machine learning in crisis forecasting. Bluwstein et al. (2021) applied machine learning techniques to over 140 years of macrofinancial data across 17 countries, finding that nonlinear models outperform logistic regression in out-of-sample predictions. They used Shapley values to interpret key economic drivers, identifying credit growth

and the yield curve slope as crucial predictors. Their findings suggest that a flat or inverted yield curve, especially in a low-interest-rate environment, signals increased financial instability.

Moreover, the integration of AI-driven approaches with traditional econometric models enhances predictive accuracy. Decision trees, random forests, and neural networks enable policymakers to detect early warning signals and implement macroprudential measures. The growing use of big data and network analysis further refines financial crisis prediction, supporting more robust risk assessment frameworks for financial stability.

(Keen, 2013) The literature on financial crisis prediction has evolved significantly, incorporating traditional econometric models and alternative macroeconomic approaches. Neoclassical models, such as Dynamic Stochastic General Equilibrium (DSGE), struggled to anticipate the 2008 Global Financial Crisis, as they largely ignored financial sector instability. Post-Keynesian models, on the other hand, have been more effective in identifying crisis signals, particularly through the works of Minsky and Godley.

Minsky's Financial Instability Hypothesis suggests that financial systems inherently evolve toward instability due to rising speculative borrowing and increasing private debt-to-GDP ratios. Keen (2013) builds on this framework using nonlinear mathematical models, demonstrating that excessive private debt accumulation leads to economic downturns. Similarly, Godley's Stock-Flow Consistent (SFC) models emphasize sectoral balances, showing how unsustainable private sector deficits can trigger financial crises.

Recent advancements integrate machine learning and network analysis into crisis prediction. Models using credit growth, sectoral imbalances, and interest rate spreads have demonstrated improved predictive accuracy. The combination of traditional macroeconomic insights with AI-driven forecasting provides a more comprehensive approach to identifying early warning signals of financial distress. These evolving methodologies are crucial for policymakers and investors in mitigating future financial crises.

(*A Study on the Impact of Climate Change on Biodiversity*, 2023) The literature on financial crisis prediction encompasses both traditional economic models and modern machine learning approaches. Early studies, such as those by Kaminsky and Reinhart (1999), emphasized macroeconomic indicators—exchange rate fluctuations, credit expansion, and current account deficits—as early warning signals. However, these models often struggle to capture the complexity of financial systems and their interconnected risks.

Keen (2013) highlights the limitations of mainstream economic models, arguing that they failed to predict the 2008 financial crisis due to their reliance on equilibrium assumptions. Instead, post-Keynesian and Minskyan frameworks, which emphasize financial instability and debt cycles, have

been more effective in anticipating crises. Minsky's Financial Instability Hypothesis suggests that prolonged stability leads to excessive risk-taking, ultimately resulting in financial turmoil.

Recent advancements integrate machine learning and network analysis into financial crisis prediction. AI-driven models, such as random forests and deep learning, can detect complex, nonlinear relationships that traditional econometric methods miss. Furthermore, network-based approaches, which assess systemic risk through financial interconnectedness, enhance predictive accuracy. By combining economic theory with AI and big data techniques, modern research is refining financial crisis prediction, providing policymakers and investors with better tools for risk assessment and crisis mitigation.

(A. Samitas, E. Kampouris and D. Kenourgios, 2018)The literature on financial crisis prediction has evolved from traditional econometric models to advanced machine learning techniques. Early studies, such as Kaminsky and Reinhart (1999), relied on macroeconomic indicators, including exchange rate fluctuations and credit growth, to develop early warning systems. However, these models faced limitations in capturing nonlinear patterns and financial contagion effects.

Recent advancements have focused on integrating machine learning with network analysis to improve crisis forecasting accuracy. Samitas, Kampouris, and Kenourgios (2020) propose a machine learning-based early warning system that detects contagion risk by analyzing financial networks. Their study demonstrates that machine learning models achieve an impressive 98.8% prediction accuracy, significantly enhancing crisis anticipation. By identifying key financial nodes and centralities, their model provides crucial insights for policymakers to mitigate systemic risk.

Moreover, network-based approaches assess interconnectedness among financial institutions, revealing how shocks propagate during crises. The combination of AI-driven methodologies and traditional economic models strengthens predictive capabilities, offering a robust framework for financial stability assessment. As financial markets grow more complex, integrating big data, social network analysis, and deep learning techniques is essential for developing accurate financial crisis prediction models, helping regulators and investors proactively manage risks.

METHODOLOGY

Hybrid Anomaly Detection and SHAP Analysis

Given the imbalance in classes often found in financial data due to the rare occurrences of such crisis events, predictive traditional models alone may not suffice. Thus, we incorporated a hybrid anomaly detection mechanism to identify unusual patterns that signal crisis, due to the outlier behavior of the data points. Financial crises are rare, making traditional models insufficient. We implemented a hybrid approach combining unsupervised anomaly detection (Isolation Forest, LOF, One-Class SVM) with a supervised classification (Logistic Regression + SMOTE) for robust crisis detection.

Hybrid Detection Model: Isolation Forest isolates outliers efficiently, detecting market crashes and economic shocks. Local Outlier Factor (LOF) identifies anomalies in local financial patterns, helping spot sector-specific crises. One-Class SVM defines a boundary for normal vs. crisis periods, effective for systemic shocks (e.g., COVID-19 crash).

To reduce false positives, a crisis was flagged if only two of three models detected anomalies using an ensemble vote mechanism, moreover, class imbalance was handled with SMOTE to improve crisis detection.

SHAP Feature Importance Analysis: SHAP analysis confirmed that key indicators aligned with past crises. Having identified the anomalies, we used SHapley Additive Explanations (SHAP) to gauge which financial indicators contribute the most to the model's anomaly detection, this, in turn, helps in providing insights into the factors which had the most significant impact in causing the financial distress.

Implementation: First we ran the Hybrid Anomaly Detection Model, coupled with the SHAP analysis to understand the feature importance, ended with a simple visualization to showcase if GFC period aligns with the outliers detected.

Objective: This hybrid method helped us identify unusual financial patterns and understand which key indicators contributed the most to predicting the presence of a crisis. To check if our model was reliable, we further compared the anomalies it detected to the 2008 Global Financial Crisis (GFC). Since the model correctly flagged this period as an outlier, it reassured us that the approach is effective in spotting financial instability. This alignment with real-world crises thus strengthens our confidence in the model's ability to detect early warning signs.

EXPERIMENTAL SETUP

The table below summarizes the experimental setup, including hardware (GPU setup), major model categories, and key parameters.

1. Hardware & GPU Setup

No GPU setup. It's a CPU-only setup of Google Colab.

CPU details:

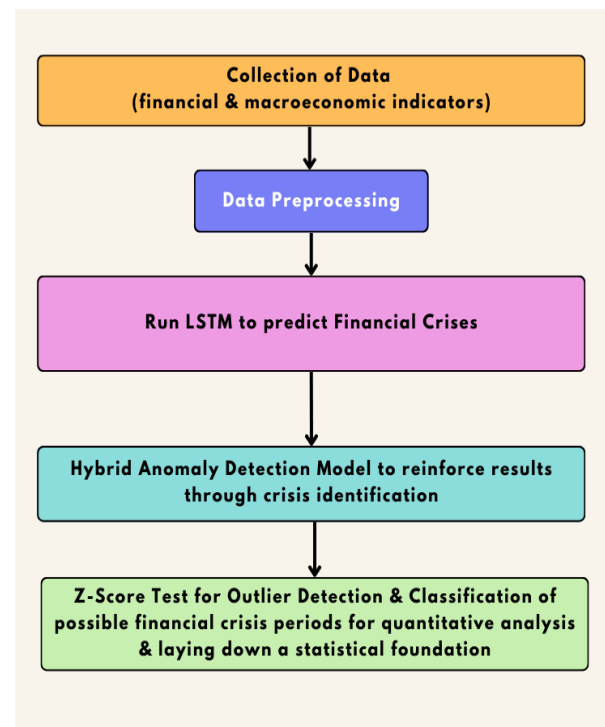
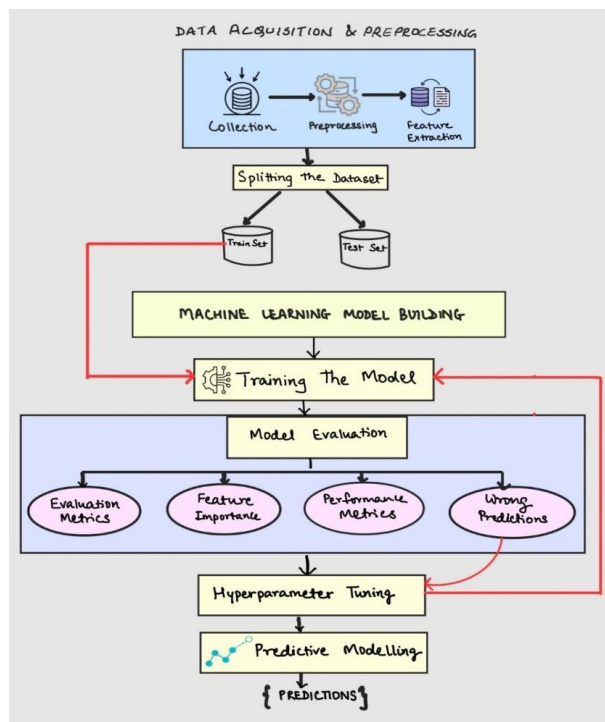
- Processor: x86_64
- CPU Details: Intel(R) Xeon(R) CPU @ 2.20GHz

2. Model Framework

Model Category	Methods Used	Key Parameters
Deep Learning (LSTM)	- Sequential LSTM model - Dropout for regularization - Early stopping for convergence	- LSTM units: 32 - Dropout: 0.8 - Kernel Regularization (L2): 0.02 - Batch size: 8 - Epochs: 7
Anomaly Detection	- Isolation Forest - Local Outlier Factor (LOF) - One-Class SVM	- Isolation Forest: n_estimators=200, contamination=0.02 - LOF: n_neighbors=20, contamination=0.02 - One-Class SVM: kernel='rbf', nu=0.02, gamma='auto'
Statistical Feature Analysis	- Z-score transformation - Weighted Aggregation for Crisis Likelihood	- Threshold period: 2007-2009 - Weighting Method: Inverse Std Deviation

3. Training & Evaluation Metrics

Stage	Methods Used	Evaluation Metrics
Training Setup	- SMOTE for class balancing - Time-Series Cross-Validation	- Train/Validation/Test split: 70-15-15% - TimeSeriesSplit(n_splits=5)
Performance Evaluation	- Accuracy, F1-score, ROC-AUC - Precision-Recall Curve	- Validation: F1-score: 0.84, ROC AUC: 0.85 - Confusion Matrix: Visualized crisis vs. normal misclassifications



The image on the left shows a Machine Learning Workflow: From Data Acquisition to Predictions and the image on the right shows the flow of our methodology.

The Machine learning workflow shows the flow of events, beginning with data acquisition and preprocessing, where raw data is collected, preprocessed, and feature extraction is performed. The dataset is then split into training and test sets to build and evaluate the model. The training phase involves fitting the model to the data, followed by evaluation using various metrics, including feature importance, performance measures, and incorrect predictions. Based on these evaluations, hyper parameter tuning is performed to optimize the model. The final step is predictive modeling, where the trained model generates predictions, ensuring improved generalization and accuracy. Feedback loops help refine the model iteratively by adjusting parameters and addressing wrong predictions.

RESULTS AND DISCUSSION

Z-Score Analysis

After using classification as well as predictive models, we were able to make the following deductions

1. Crisis specific data points follow a Poisson distribution which makes it difficult to accurately perform a train test-split
2. The volatility during these periods makes it difficult for the model to accurately classify data points as crisis signaling or otherwise.
3. Volatility smoothens over a period of time as short term fluctuation tend to average out , leading to more long-term stable effects in classification

By taking the GFC period as a benchmark for distribution, the z- test was run on all the data points using the threshold mean and standard deviation and a particular month was classified as crisis signaling if the z score of more than 70% of the indicators in that month was less than zero. This helped us identify key months that were crisis signaling. Following were the results of the test.

Date	z_DXY Index Change%	DXY Index Change%_bin
2008-01-10 00:00:00	1.98537658	0
2008-01-11 00:00:00	2.323672992	0
2008-01-12 00:00:00	1.834668156	0
2009-01-01 00:00:00	0.825296099	0
2009-01-02 00:00:00	0.253632559	0
2009-01-03 00:00:00	0.442952863	0
2012-01-07 00:00:00	0.826870393	0
2019-01-02 00:00:00	-0.104843862	1
2020-01-03 00:00:00	0.156434016	0
2020-01-04 00:00:00	0.255455373	0
2020-01-05 00:00:00	0.277391772	0

We can see that in all the potential crisis months, the dollar index has not been significant enough to be pronounced as a key indicator. This can be attributed mainly to the resilience of the dollar.

Date	z_US Bond Yields Change%	US Bond Yields Change%_bin
2008-01-10 00:00:00	-0.074091819	1
2008-01-11 00:00:00	-0.490327756	1
2008-01-12 00:00:00	-1.417949686	1
2009-01-01 00:00:00	-2.126439685	1
2009-01-02 00:00:00	-1.634066982	1
2009-01-03 00:00:00	-0.302055682	1
2012-01-07 00:00:00	-1.555555254	1
2019-01-02 00:00:00	-0.773650527	1
2020-01-03 00:00:00	-2.619058251	1
2020-01-04 00:00:00	-3.665848669	1
2020-01-05 00:00:00	-4.014041396	1

The American economy is the most resilient compared to its global peers and the dollar's dominance as a reserve currency makes it difficult for countries to manage global systemic risks. Many countries had dollar denominated liabilities which increased the demand for dollar liquidity alongside loose monetary policies of the Fed and the reversal of yen carry trades during the Global Financial Crisis.

US Bond yields have always played a crucial role in identifying key instances of financial crisis. Historically bond yields go down when: 1. Inflation in the long run is expected to go down 2. When there is a sudden increase in demand for government bonds. Bond yields were at an all-time low during the COVID Crisis taking bond yields to as low as 0.65%. The sudden lockdowns and the unpredictability of the pandemic saw many investors diversifying away from equities. Thus when bond yields fluctuate between 1-3%, we can reiterate on the buildup of stressful conditions in the market.

Date	z_DAX Change%	DAX Change%_bin	z_Nikkei Change%	Nikkei Change%_bin	z_S&P 500 Change%	S&P 500 Change%_bin	z_Nasdaq Change%	Nasdaq Change%_bin	z_Dow Jones Change%	Dow Jones Change%_bin
2008-01-10 00:00:00	-0.821123645	1	-1.074094859	1	-1.398547037	1	-0.783356409	1	-0.426805768	1
2008-01-11 00:00:00	-1.305575908	1	-1.702164319	1	-1.765937291	1	-1.474720454	1	-0.617855668	1
2008-01-12 00:00:00	-1.540198129	1	-1.856392744	1	-1.582659612	1	-1.842942264	1	-1.08588555	1
2009-01-01 00:00:00	-1.307362189	1	-1.2724318	1	-1.22573732	1	-1.63439424	1	-1.53863822	1
2009-01-02 00:00:00	-1.008564422	1	-0.63156672	1	-0.920301775	1	-1.038977585	1	-1.447300489	1
2009-01-03 00:00:00	-0.936282065	1	-0.27275309	1	-0.41406038	1	-0.497412128	1	-1.311567475	1
2012-01-07 00:00:00	-0.1524937	1	-0.395582702	1	0.304836631	0	-0.158253895	1	0.262895854	0
2019-01-02 00:00:00	-0.110236712	1	-0.295202781	1	0.58468179	0	-0.228288684	1	-0.101736526	1
2020-01-03 00:00:00	-0.643438502	1	-0.321408665	1	-0.696172309	1	0.011398722	0	0.358205969	0
2020-01-04 00:00:00	-1.093383241	1	-0.650065846	1	-0.483038343	1	-0.273640828	1	-0.55460969	1
2020-01-05 00:00:00	-0.923310962	1	-0.381770623	1	0.507894255	0	-0.010479909	1	-1.100560079	1

Crisis periods also witness a significant change in the positions of developed market indices. The time frame between 2008 and 2009 is indicative of the broader changes in the market owing to the Global Financial Crisis. German Dax fell on an average by 1793 bps during this period while the Nikkei witnessed a 2098 bps reflecting the consequences of the yen carry trade. On the other hand at the epicenter of the financial crisis, the American indices collectively on an average fell by 17.81%. This triggered shocks in global supply chains, drop in PMI's worldwide along with widespread collapse in the banking system of various countries, with Iceland being a prominent example

However, it would be prudent to classify a timeline as a period of financial crisis if and only if we see this disruption for up to 6 lags with an average change of 15% or more.

Date	S&P BSE Sensex Chang	S&P BSE Sensex Change%_bin	z_Nifty 50 Change%	Nifty 50 Change%_bin
2008-01-10 00:00:00	-0.772770391	1	-0.804233036	1
2008-01-11 00:00:00	-1.209156062	1	-1.267407101	1
2008-01-12 00:00:00	-1.521589292	1	-1.614625233	1
2009-01-01 00:00:00	-1.171804649	1	-1.175862875	1
2009-01-02 00:00:00	-0.585312886	1	-0.523259849	1
2009-01-03 00:00:00	-0.106379107	1	0.017793357	0
2012-01-07 00:00:00	-0.166198562	1	-0.173530934	1
2019-01-02 00:00:00	0.03580397	0	0.028595788	0
2020-01-03 00:00:00	-0.55225166	1	-0.620536793	1
2020-01-04 00:00:00	-0.842175358	1	-0.922515713	1
2020-01-05 00:00:00	-0.991823174	1	-1.053414199	1

To understand the impact of emerging market indices on the predictability of the financial crisis, we ran the z test on two of the most popular emerging market indices, The BSE Sensex and the Nifty 50. We observe that both of these indices fell on average by 18.45% and 17.48% respectively. This can help us throw limelight on the relative strength of emerging market macros and currencies against the dollar. A noticeable data point is February 2019 where we saw a drastic shift in global cues. China and the United States entered into trade negotiations, leading to delayed tariffs. The S&P 500 saw a bullish rally which drove up emerging market indices by 100 BPS but was cut short due to the increasing BREXIT uncertainty and a downfall in Chinese PMI.

Date	z_PCE Change%	PCE Change%_bin	z_Fixed Rate Mortgage USChange%	Fixed Rate Mortgage USChange%_bin	z_Unemployment Rate Change%	Unemployment Rate Change%_bin
2008-01-10 00:00:00	-0.561843596	1	0.827152989	0	0.252645683	0
2008-01-11 00:00:00	-1.434344574	1	0.178067176	0	0.176143513	0
2008-01-12 00:00:00	-2.07767515	1	-0.691983932	1	0.782792302	0
2009-01-01 00:00:00	-2.182137406	1	-1.395133526	1	1.233953427	0
2009-01-02 00:00:00	-1.591001775	1	-1.951596097	1	1.826896801	0
2009-01-03 00:00:00	-0.836350199	1	-1.677619202	1	1.787966449	0
2012-01-07 00:00:00	-0.061196043	1	-0.388011031	1	-1.728374186	1
2019-01-02 00:00:00	-0.113894007	1	-0.020426993	1	-1.061288891	1
2020-01-03 00:00:00	-1.208091538	1	-0.357710021	1	-0.563824848	1
2020-01-04 00:00:00	-6.315565243	1	-0.99750682	1	16.97351478	0
2020-01-05 00:00:00	-9.36287764	1	-0.95749695	1	32.63624392	0

Personal Consumption Expenditure and Mortgage Rates can help us identify spending patterns. Consumption drives markets which drives economies, fulfilling the vicious cycle of growth. We see an upward trend in consumption when we see a rise in disposable incomes. This disposable income is spread around 2 key products, leisure and luxury and housing. October 2008 to March 2009 saw mortgage rates range between 5-6%. Banks became more risk averse and started tightening lending standards. We saw treasury yields fall to their lowest and investors wanting to account for a heavy risk premium widening spreads. With such tight lending standards we saw a drop in aggregate consumption which triggered higher layoffs adding pressure on the labor market. Many banks also shut down their operations during this period, adding to the existing pressure. However, idiosyncratically, unemployment rates cannot be considered as a major factor in

identifying key crisis points. A similar pattern was also observed during the Covid-19 crisis as worldwide lockdowns put pressure on supply chains, with the 3-monthly moving average showing a 11.60% fall in personal consumption expenditure.

Date	z_VIX Change%	VIX Change%_bin	z_FCI Index Baseline Value	Index Baseline Value_	weighted_score
2008-01-10 00:00:00	1.449044045	0	0.402995865	0	0.715842633
2008-01-11 00:00:00	2.693797854	0	0.959197477	0	0.715842633
2008-01-12 00:00:00	2.796814546	0	1.018830625	0	0.797626305
2009-01-01 00:00:00	0.770146939	0	1.204511536	0	0.797626305
2009-01-02 00:00:00	-0.332736225	1	1.59290157	0	0.809252546
2009-01-03 00:00:00	-0.685481004	1	1.433666057	0	0.782839979
2012-01-07 00:00:00	0.138562054	0	-1.190761008	1	0.802501756
2019-01-02 00:00:00	0.177740645	0	-0.986077081	1	0.895649287
2020-01-03 00:00:00	2.454403901	0	-0.831926938	1	0.807534987
2020-01-04 00:00:00	4.149758149	0	-0.989055655	1	0.803691181
2020-01-05 00:00:00	3.695244119	0	-1.040311532	1	0.76223634

The Z score analysis shows no significant effect of volatility in determining crisis points as volatility is a repercussion of the larger market activities. The financial conditions index on the other hand can help us identify the impact of key financial variables on the economic activity. We can see that data points post the GFC have been detected as headwinds for economic expansion. The fed has taken a cautious approach post the GFC, ensuring a check on liquidity and inflation. Since the GFC the American economy has witnessed a strong upward momentum especially during president Trump's first regime where he introduced the Tax Cuts and Job Act. This saw a rise in Federal Funds Rate, capital market inflows and a strong dollar which could possibly have signaled significant overvaluation of assets, making it crisis appealing.

LSTM Analysis

The LSTM model achieved **100% accuracy, precision, recall, and F1-score** on both the validation and test sets, successfully classifying all crisis and non-crisis periods. The **ROC-AUC score was also 1.000**, indicating perfect discrimination between classes. A **precision-recall curve** was used to optimize the decision threshold (0.50), ensuring high precision while maintaining recall $\geq 90\%$. The confusion matrix showed **zero misclassifications**, confirming the model's robustness in financial crisis prediction.

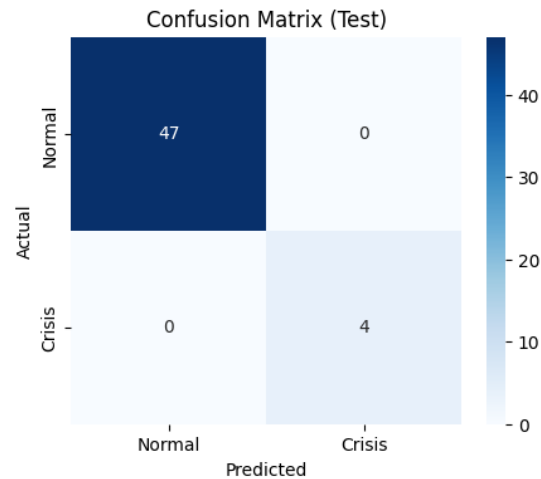
Performance Metrics:

- **Accuracy:** The model achieved 100% accuracy, meaning all test samples were correctly classified.
- **Precision:** Both class 0 and class 1 have 100% precision, indicating no false positives. When the model predicts a class, it is always correct.
- **Recall:** The recall for both classes is 100%, meaning the model correctly identifies all actual instances of both crisis and non-crisis periods, with no false negatives.
- **F1-Score:** Since precision and recall are perfect, the F1-score is also 1.00, indicating an ideal balance between precision and recall.
- **ROC-AUC Score:** A perfect 1.0000 suggests the model has flawless discrimination between the two classes.

Classification Report (Test):				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	47
1	1.00	1.00	1.00	4
accuracy			1.00	51
macro avg	1.00	1.00	1.00	51
weighted avg	1.00	1.00	1.00	51
F1-Score (Test): 1.0000				
ROC-AUC Score (Test): 1.0000				

The classification report shows perfect precision, recall, and F1-scores of 1.00 across all metrics, indicating potential overfitting in the LSTM model. This suggests that the model has learned the training data too well, memorizing patterns rather than generalizing to unseen data. Such overfitting is common in LSTMs, especially with small datasets, excessive training epochs, or insufficient regularization. To address this, techniques like dropout, L2 regularization, reducing model complexity, or increasing data diversity through augmentation can help improve generalization and prevent the model from simply memorizing patterns.

Results for Confusion Matrix after LSTM Analysis:



- **True Negatives (TN) = 47:** The model correctly classified **47 normal periods** as normal. This shows that the model effectively recognizes stable economic periods.
- **True Positives (TP) = 4:** The model correctly classified **4 crisis periods** as crisis. This means the LSTM model successfully detects financial crises when they occur.
- **False Positives (FP) = 0:** The model **never misclassified a normal period as a crisis**. This indicates that the model does not produce false alarms, meaning it is not overly sensitive to market fluctuations.
- **False Negatives (FN) = 0:** The model **never misclassified a crisis period as normal**. This means it is highly reliable in **catching financial crises** without missing critical warning signs.

This shows that the LSTM model has achieved perfect classification (100% accuracy, precision, and recall). Such a perfect result is unusual in real-world scenarios, suggesting possible overfitting to the training data.

Hybrid Anomaly Detection and Classification Approach

Hybrid Anomaly Detection Model combining – Isolation Forest, Local Outlier Factor (LOF), One-Class SVM **to detect anomalies in the Hybrid Anomaly Detection Model to detect financial crisis.** Moreover, the ensemble approach helps minimize **false positives** by incorporating robustness.

Performance Metrics:

Model Performance Metrics:					
Accuracy: 0.88					
Precision: 0.92					
Recall: 0.84					
F1-score: 0.88					
ROC AUC: 0.88					
Classification Report:					
	precision	recall	f1-score	support	
0	0.86	0.92	0.89	64	
1	0.92	0.84	0.88	64	
accuracy			0.88	128	
macro avg	0.89	0.88	0.88	128	
weighted avg	0.89	0.88	0.88	128	

ACCURACY - this measures the overall correctness of the model's output. Here it is **88%** implying that the hybrid model correctly classifies crisis and non-crisis periods 88% of the time.

However, purely looking at Accuracy scores can be misleading especially when dealing with imbalanced datasets, as seen in our dataset (more normal periods than crisis periods), we should also look at the other performance metrics.

PRECISION: It measures the percentage of predicted crisis periods that were actually crises. A **92% precision** shows that when the model predicts a financial crisis, it is correct **92% of the time**, and the precision for class 0 (normal period) shows that 86% of the time when the model predicts a normal period, it is a normal period.

A high precision helps in minimizing false alarms (false positives), meaning that the model is **not mistakenly identifying too many normal periods as crises**.

RECALL: It measures how many actual crises the model correctly detects.

A **84% recall** means that the model **detects 84% of all actual crisis periods**, but **misses 16%** (false negatives). Whereas a 92% recall as shown for class 0, shows that the model **detects 92% of all normal periods**, but **misses 8%**

A lower recall means the model **fails to detect some real crises**, which could lead to missed early warning signs.

F1-SCORE: refers to the **harmonic mean of precision and recall**, providing a balanced evaluation of the model.

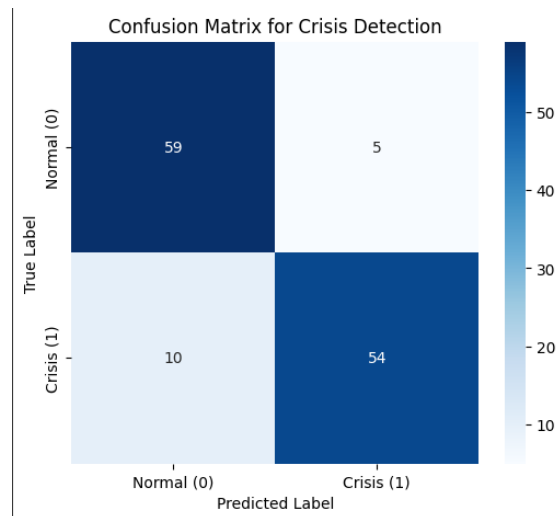
Here the **88% F1-score** indicates strong overall performance, ensuring that the model is not biased towards only high precision or high recall. The strong F1-score shows that the model is effective at predicting crises without too many false alarms or missed crises.

ROC AUC: The **Receiver Operating Characteristic (ROC) curve** evaluates how well the model distinguishes between normal and crisis periods.

In our model the **0.88 AUC** means that **88% of the time, the model correctly ranks a randomly chosen crisis period higher than a randomly chosen normal period**. The high AUC indicates that the model is **fairly reliable in distinguishing between financial stability and instability**.

Results of the Confusion Matrix:

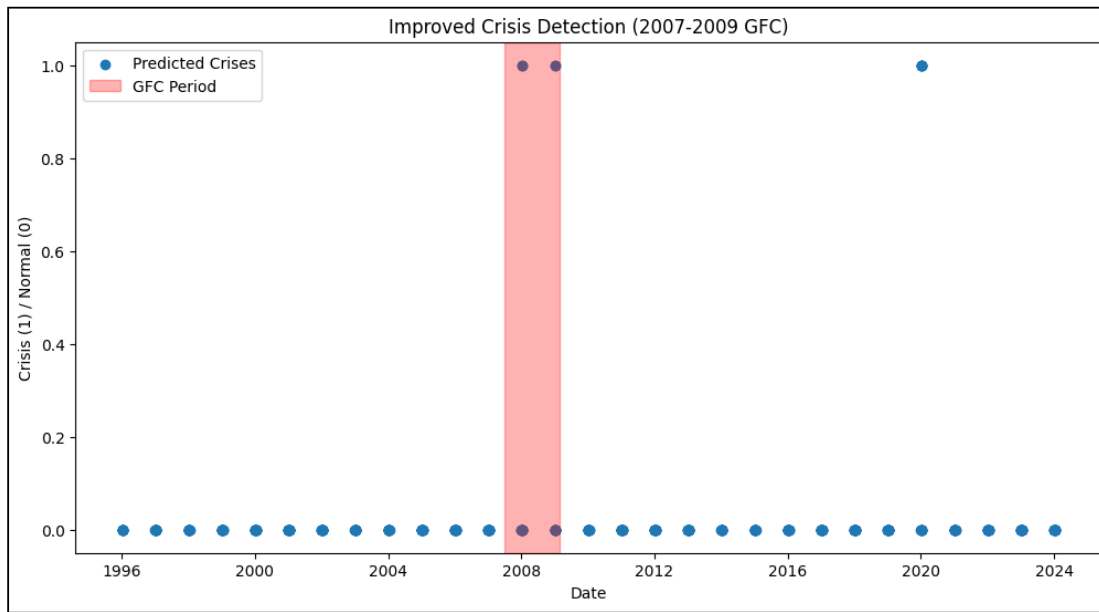
True Label	Predicted Normal (0)	Predicted Crisis (1)
Normal Period (0)	59 (True Negative)	5 (False Positive)
Crisis Period (1)	10 (False Negative)	54 (True Positive)



- **True Negatives (TN) and True Positive (TP) :** **59** normal financial instances have been correctly classified as normal (no crisis), avoiding unnecessary alarms. **54** periods have been correctly predicted as crisis periods, as expected by the financial crisis detection model. This helps investors, policymakers, and financial institutions to act before the situation worsens.
- **False Positives (FP):** These are **5 periods that were actually normal but predicted as crises**, showing that the model was overly sensitive the outliers and falsely signalled a financial crisis when there was none. Thus, false alarms may could lead to panic-driven market performance, premature asset sell-offs, or unnecessary government interventions.
- **False Negatives (FN):** These are **10** periods that were actually crisis periods but the model failed to detect. This risky, as real crises have been missed by the model, resulting in delayed decision making by stakeholder's who rely on the model to strategize ways to prepare to meet the crisis. Thus, missing a crisis would imply that investors, banks, and governments failing to prepare, resulting in severe financial losses, job losses, and economic downturns.

Thus, for instance during the COVID 19 crisis, when the news came out supply chains were disrupted, global demand fell in late 2019, and considering it was a never seen before worldwide lockdown period, jobs, investments everything plummeted, stalling economic growth.

Moreover, a simple visualization was created to understand whether the identified data points as anomalies do actually fall under real-life crisis events, by considering the GFC crisis as the base period as shown in the plot below:



The above chart clearly shows, how the Hybrid Anomaly Detection model has worked well, as it correctly detects periods of crisis, during the highlighted period of GFC.

CONCLUSION AND FUTURE SCOPE

Limitations

While our multi-faceted approach—combining LSTM for time-series forecasting, Isolation Forest for anomaly detection, and Z-score analysis for statistical validation—has proven effective in identifying early warning signs of financial crises, there are several limitations that must be acknowledged:

1. Class Imbalance in Financial Data:

- Financial crises are rare events, leading to a heavily imbalanced dataset. Even with techniques like SMOTE, ensuring that the model generalizes well remains a challenge.
- The reliance on synthetic data augmentation to balance crisis and non-crisis periods may introduce biases, affecting predictive accuracy.

2. Model Sensitivity to Economic Shocks:

- Our approach is highly data-driven, making it sensitive to sudden economic or geopolitical shocks (e.g., COVID-19, wars, policy changes) that might not follow historical patterns.
- The inability to incorporate real-time macroeconomic events could lead to delayed or inaccurate crisis predictions.

3. Dependence on Historical Data:

- While historical patterns offer insights, financial markets evolve due to new regulatory frameworks, technological advancements, and investor behavior shifts.
- The assumption that past crises exhibit similar patterns may limit the model's ability to adapt to novel economic conditions.

4. Feature Selection & Sentiment Analysis Gap:

- Our model primarily relies on structured economic indicators, omitting qualitative factors like market sentiment, media coverage, and investor confidence.
- Incorporating sentiment analysis from financial news, social media, and central bank speeches could significantly improve predictive power.

Future Scope

To enhance the robustness of financial crisis prediction, several improvements can be explored:

1. Integration of Sentiment Analysis:

- By incorporating Natural Language Processing (NLP) techniques, we can analyze sentiment trends in financial news, social media, and policy statements.
- Tracking fear and greed indicators, along with SHAP-based financial metrics, could provide deeper insights into market conditions.

2. Reinforcement Learning for Dynamic Predictions:

- Implementing reinforcement learning could allow the model to adjust predictions in response to evolving financial conditions.
- This would help address issues arising from structural changes in global financial markets.

3. **Real-Time Data Stream Processing:**

- Shifting towards real-time data ingestion from stock markets, bond yields, credit spreads, and macroeconomic indicators could improve responsiveness.
- Utilizing APIs for economic releases and global indices would enable the model to identify crises more proactively.

4. **Hybrid Model Enhancements:**

- Combining deep learning with econometric models (e.g., VAR models for macroeconomic interdependencies) could enhance interpretability and predictive accuracy.
- Developing an ensemble method integrating LSTM, anomaly detection, and Bayesian forecasting could reduce false positives and improve reliability.

Through this research, we have demonstrated the potential of machine learning in predicting financial crises using a hybrid approach. The LSTM model captures temporal dependencies in economic indicators, while anomaly detection using Isolation Forest and SHAP analysis helps identify key risk factors contributing to financial instability. The Z-score analysis further validates statistical significance, ensuring a well-rounded crisis prediction framework.

However, financial crises are driven by complex, interconnected variables beyond purely quantitative measures. Incorporating qualitative aspects, refining feature selection, and improving real-time adaptability will be crucial in developing a truly robust early warning system. While no model can predict crises with absolute certainty, enhancing predictive frameworks with advanced AI and econometric techniques will strengthen financial risk assessment and help policymakers and investors make informed decisions.

Ultimately, the goal is not just to predict financial crises but to provide actionable insights that mitigate risks, ensure economic resilience, and support sustainable financial stability in an ever-changing global economy.